Privacy-preserving Federated Learning (PPFL) leveraging TinyML framework for resource-constrained IoT devices under multimode communication protocols

Fahim Hafiz
Lecturer, CSE
United International University

Dr. Mirza Rasheduzzaman
Associate Professor, EEE
BRAC University

Dr. Mohammad Shahriar Rahman
Professor, CSE
United International University

Mehedi Hasan Shawon
Lecturer, EEE
BRAC University

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Introduction

Integrating Artificial Intelligence (AI) into the decision-making processes of the Internet of Things (IoT) aims to create systems that are not only *connected* but also *intelligent and autonomous*.

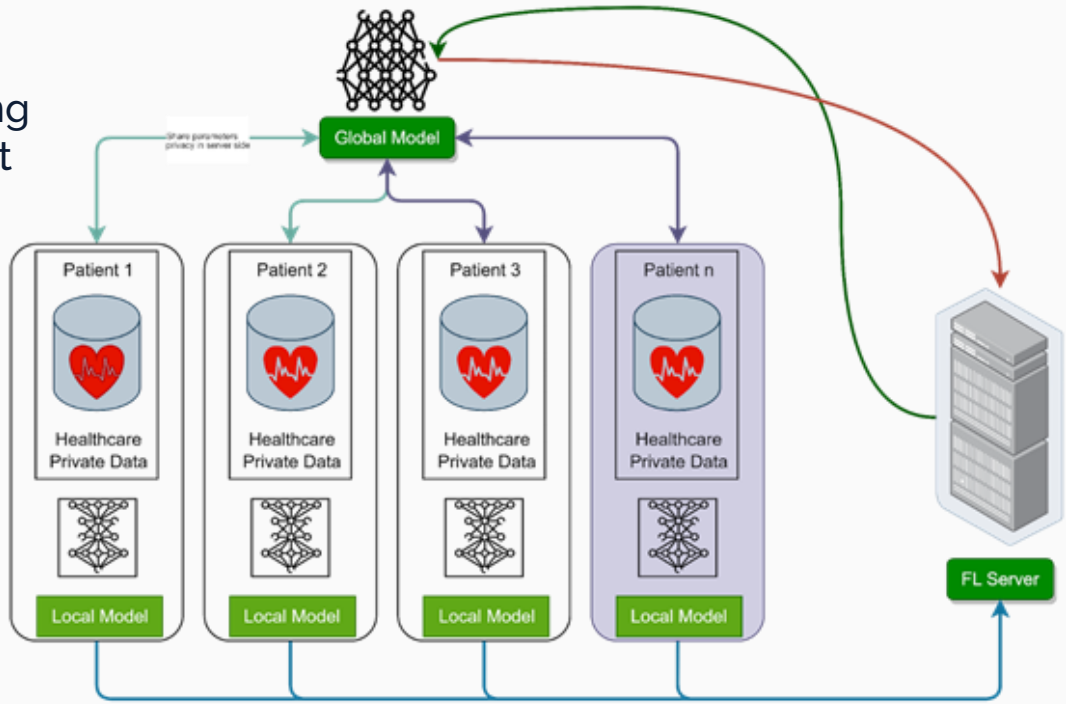
- Traditionally IoT devices generate continuous, real-time data for central predictive analysis
- This centralization can lead to vulnerabilities, such as data breaches and unauthorized access, potentially violating regulatory guidelines like GDPR, HIPAA, and PDPA.
- We propose a modular, privacy-preserving framework using:
  - TinyML for edge computing , optimized for resource-constrained IoT environments
  - Federated Learning for decentralized model training
  - Multi-mode communication for resilient connectivity

Federated Learning

Decentralized Yet Coordinated.

A method to train a machine learning model across many devices without moving the data.

- Only model updates (weights/gradients) sent to a central aggregator
- Global model updated using algorithms like FedAvg
- No raw sensor data leaves the device
- Supports privacy, regulatory compliance, and bandwidth efficiency



Why Combine Federated Learning with TinyML

- TinyML allows models to run on tiny, low-power IoT devices.
- FL allows training to happen privately on each device.
- Together, they enable private learning directly on resource-constrained nodes.
- We avoid centralized cloud storage of sensitive data.

Literature Review – Recent Advances & Gaps

Study	Focus Area	Key Contributions	Limitations
Abadade et al. (2023) [1]	TinyML	Review of low-power ML on MCUs; highlights success in healthcare & agriculture	Lacks integration with privacy-preserving training
Lazaros et al. (2024) [2]	Federated Learning	Discusses decentralized model training, communication optimization	Focused more on theory; minimal IoT hardware validation
Dritsas & Trigka (2025) [3]	FL for IoT	Evaluates real-world FL deployments with resource constraints	Does not address secure model sharing or multi-mode comms

02

Objectives of the project

Project Goals



TinyML (On-Device Intelligence)

- Deploy a modified TinyML framework on low-power IoT devices for on-device training and real-time processing of sensor data



Federated Learning for Decentralization

- Implement Federated Learning on edge devices with synchronized updates across nodes despite constraints in wireless communication



Multi-mode Communication

- Develop a hierarchical network topology for nodes to dynamically switch between protocols for adaptive & resilient transmission



Cybersecurity Integration

- Incorporate cybersecurity measures for securing the communication protocols, data storage, and for detecting real-time attacks

03

Project Description

Methodology

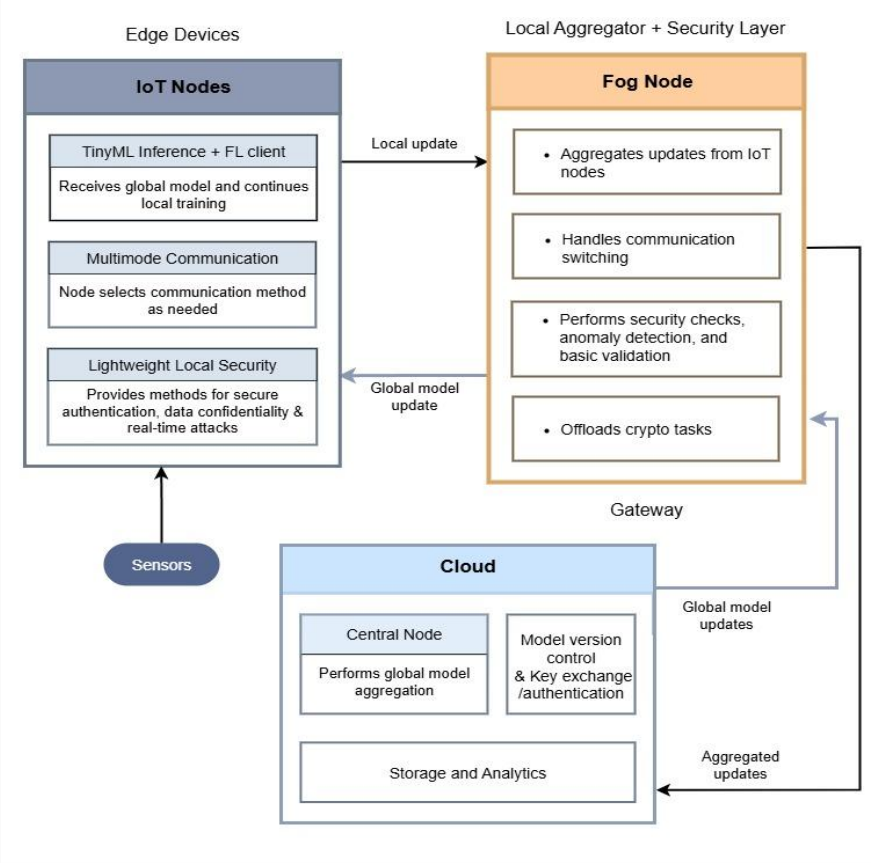
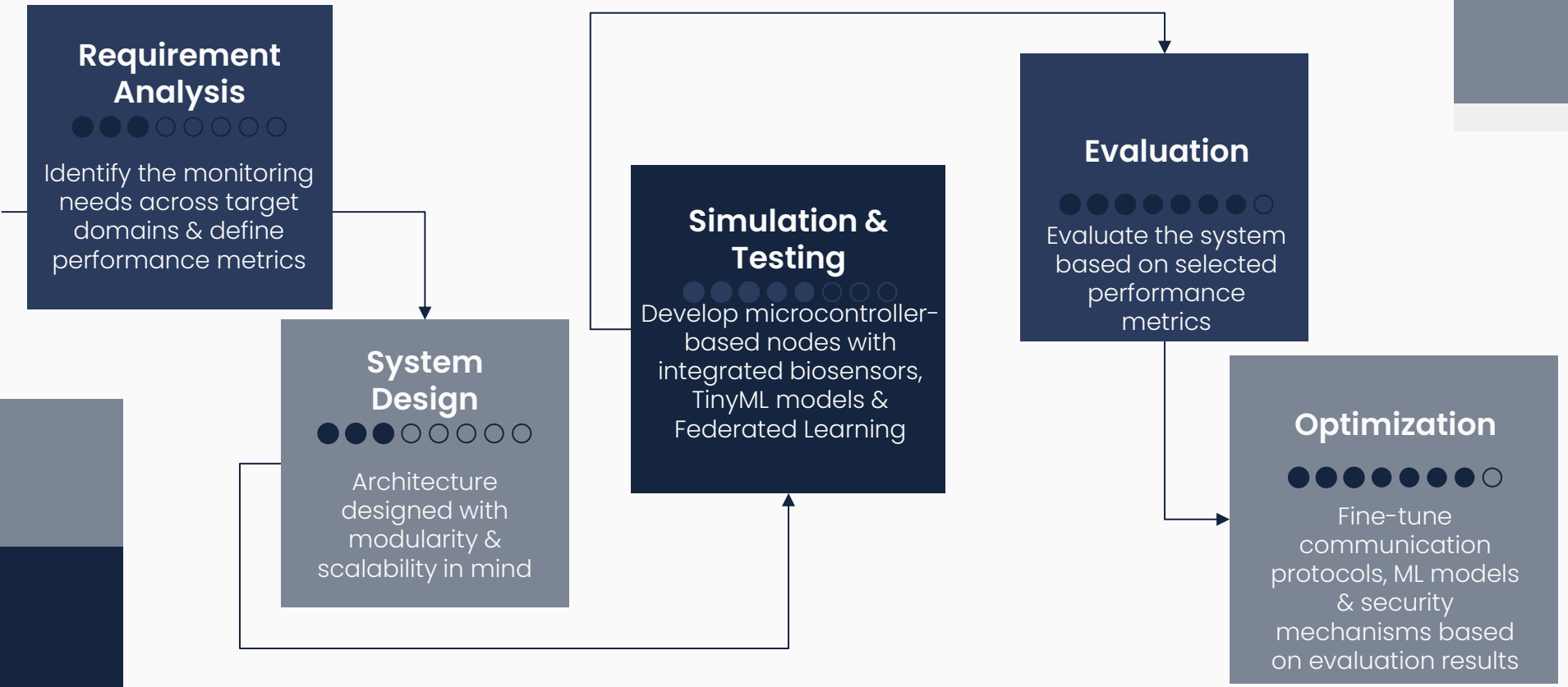
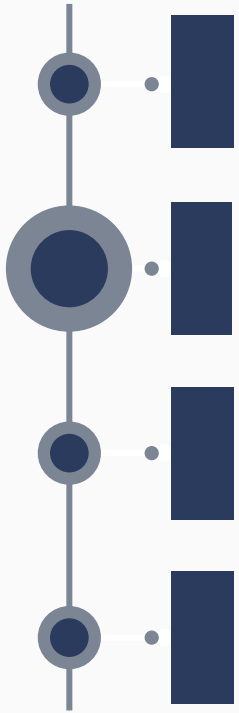


Fig: TinyML-FL Framework with Edge-Fog-Cloud Collaboration

Methodology Phases



Methodology Phases

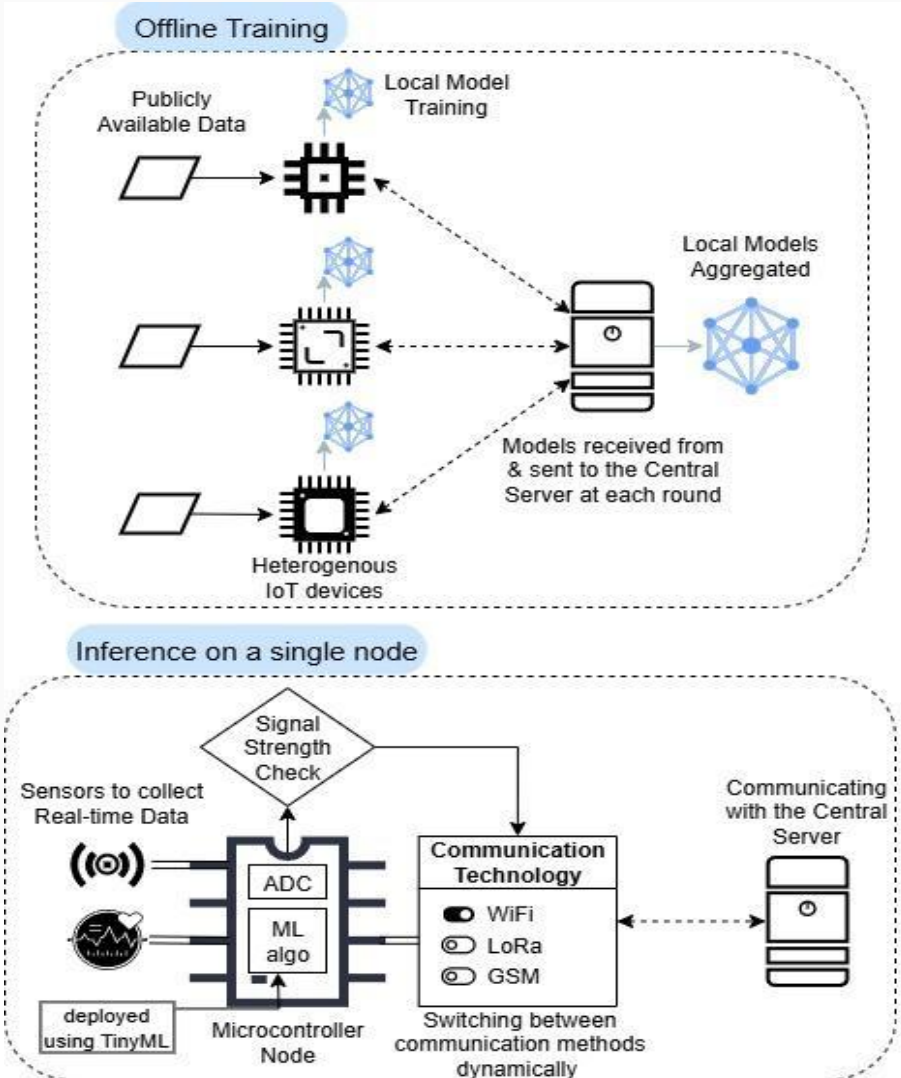


The architecture must contain these core components:

- TinyML models on microcontroller-based nodes for local real-time processing of sensor data
- Federated learning for decentralization, along with framework development for resource allocation
- Support for diverse communication technologies to securely relay aggregated data to the cloud
- Robust security measures such as AES-256 & RSA encryption for data security, Differential Privacy & Homomorphic encryption for data protection during training and communication

Validation Strategy

- No real patient data will be collected in this project initially.
- We will use publicly available datasets (e.g., ECG, PPG from PhysioNet, UCI)
  - To pre-train machine learning models,
  - To simulate federated learning (FL) across multiple devices,
  - To test inference accuracy on real hardware.
- Complex FL behaviors and communication patterns will be simulated using tools like Flower, TFF, and NS3.
- Real IoT devices will:
  - Run TinyML models locally,
  - Simulate FL update exchange,
  - Demonstrate communication switching (WiFi, LoRa, GSM) based on real-world conditions.



Performance Metrics

Model Performance

- Accuracy
- Inference time
- Learning convergence
- Efficiency

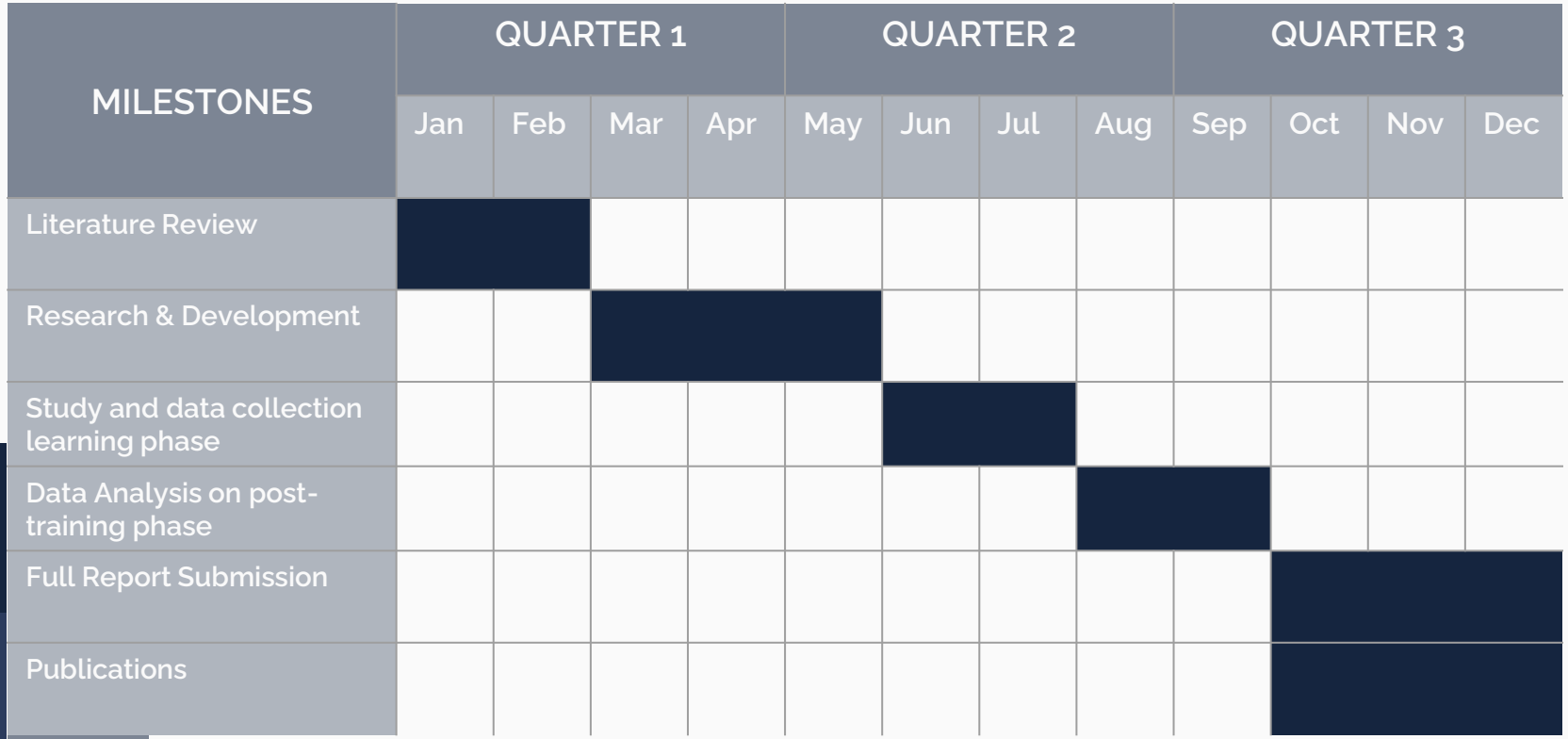
Communication Efficiency

- Data throughput
- Coverage
- Power Consumption

Security Robustness

- Resistance to attacks
- Encryption strength
- Authentication reliability

Project roadmap



Expected Outcomes

- Working IoT nodes with pre-trained TinyML models doing live inference.
- A federated learning simulation across clients using public datasets.
- A demonstration of secure model updates and communication switching.
- A modular and reusable system architecture for real-world use.

Conclusion

- **Key Impact**: Privacy-preserving, real-time edge intelligence
- **Scalability**: Modular design for healthcare, agriculture, industry
- **Feasibility**: Simulation + lightweight deployment
- **Alignment**: Supports SDGs 9, 11, 16

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